Exchange Rate Regimes in Production Networks

A Small Open Economy Extension of Rubbo (2024)

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Building on: Rubbo (2024) *Networks, Phillips Curves, and Monetary Policy*

Outline

Intro

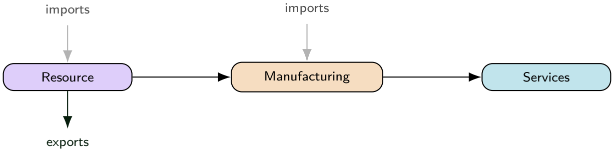
Model

Results

Conclusion

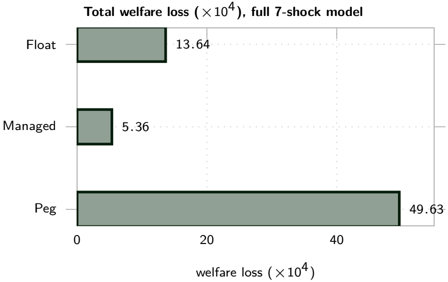
Motivation

- Network Phillips curves (Rubbo 2024, La'O–Tahbaz-Salehi 2022): **closed economy**, unique DC index
- Open-economy NK (Galí–Monacelli 2005): FX channel, **no network**
- **This paper: both together** — multi-sector network + FX



Questions: Optimal FX regime? Does the DC index survive a 2nd cost-push channel?

Headline Result



Units: consumption-equivalent loss, $\times 10^{-4}$ throughout.

- **Managed float dominates** both corners
- **Peg dominated**: zero autonomy, UIP shock passes straight through
- **Driver:** persistence (near-unit-root NFA), not volatility

Literature

Network side (closed economy):

- **Rubbo (2024)** — Calvo network $\Rightarrow$ unique DC index (1 cost-push channel)
- **La'O & Tahbaz-Salehi (2022)** — optimal MP in production networks
- **Pasten, Schoenle & Weber (2020)** — sectoral rigidity heterogeneity drives MP transmission

Open-economy side (no network):

- **Galí & Monacelli (2005)** — 1-sector SOE: domestic-inflation targeting $\approx$ optimal, peg dominated (ToT-adjustment channel)
- **Fanelli & Straub (2021)** — FX intervention as a *second* instrument, separate from i_t
- **Schmitt-Grohé & Uribe (2003)** — debt-elastic premium to stationarize NFA; flags 1st-order welfare risk
- **Gopinath & Itskhoki (2022); Amiti et al. (2019); Auer, Levchenko & Sauré (2019)** — pass-through, dominant currency, IO spillovers

Gap: network + open economy + FX together — no one has this.

- Same ranking as Galí–Monacelli, **different channel** (risk-premium, not ToT)
- Managed's dominance echoes Fanelli–Straub's two-instrument logic
- Sector heterogeneity = FX analogue of Pasten-Schoenle-Weber

This Paper

- Extend Rubbo → SOE: imported inputs Ω^F, UIP, debt-elastic premium
- Ask: optimal FX regime? DC index survive with 2 cost-push channels?
- **Not just Galí-Monacelli with sectors**: ranking survives, but the mechanism (risk-premium) and sector heterogeneity are new
- Calibrate: 3-sector oil exporter; solve **nonlinear** model directly (not hand-linearized)

Model: Households

$$\max\ \mathbb{E}_0 \sum_t \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right] \qquad \text{s.t.} \qquad P_t^C C_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i^*)\left[1-\psi(\hat{b}_{t-1}^* - \bar{b}^*)\right] S_t B_{t-1}^*$$

$$C_t^{-\gamma} = \beta(1+i_t)\mathbb{E}_t\left[C_{t+1}^{-\gamma}/\Pi_{t+1}^C\right], \qquad W_t/P_t^C = C_t^\gamma L_t^\varphi$$

$$\text{UIP (derived):} \quad (1+i_t) = (1+i^*)\left[1-\psi(\hat{b}_t^* - \bar{b}^*)\right] \mathbb{E}_t[S_{t+1}/S_t]$$

C, L: consumption, labor — β, γ, φ: discount, RRA, inv. Frisch — W, i: wage, dom. rate — S, i^*: ER, foreign rate

ψ: debt-elastic premium — B^*: foreign bond — P^C: CPI — Π_t: profits (rebated)

Model: Firms — Technology and Cost

$$Y_{it} = A_{it}\, L_{it}^{\alpha_i} \prod_j X_{ijt}^{\Omega_{ij}^H}\, M_{it}^{\Omega_i^F} \qquad \Rightarrow \qquad MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} \left(S_t P_t^{F*}\right)^{\Omega_i^F}}{A_{it}}$$

$$W_t L_{it} = \alpha_i MC_{it} Y_{it}, \qquad P_{jt} X_{ijt} = \Omega_{ij}^H MC_{it} Y_{it}, \qquad S_t P_t^{F*} M_{it} = \Omega_i^F MC_{it} Y_{it}$$

Y, A: output, TFP L, X, M: labor, dom./imp. inputs Ω_{ij}^H: dom. IO share Ω_i^F: import cost share
α_i: labor share MC_{it}: nominal marg. cost P^{F*}: foreign import price

Key: MC_{it} moves with both A_{it} (TFP) and S_t (via M_{it}) — the two cost-push channels.

Model: Firms — Calvo Pricing

Reset price (probability δ_i resets, $1-\delta_i$ keeps last price):

$$X1_{it} = \widetilde{MC}_{it} Y_{it} + (1-\delta_i)\beta\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}\Pi_{i,t+1}^{\varepsilon} X1_{i,t+1}, \qquad X2_{it} = Y_{it} + (1-\delta_i)\beta\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}\Pi_{i,t+1}^{\varepsilon-1} X2_{i,t+1}$$

$$P_{it}^{\#}/P_{it} = \frac{\varepsilon}{\varepsilon-1} X1_{it}/X2_{it}, \qquad 1 = (1-\delta_i)\Pi_{it}^{\varepsilon-1} + \delta_i\left(P_{it}^{\#}/P_{it}\right)^{1-\varepsilon}$$

δ_i: Calvo reset prob. (primitive) $\quad \hat{\delta}_i$: derived slope $\quad P_{it}^{\#}$: reset price $\quad \varepsilon$: elasticity (markup $\varepsilon/(\varepsilon-1)$)

$\hat{\delta}_i = \dfrac{\delta_i(1-\beta(1-\delta_i))}{1-\beta\delta_i(1-\delta_i)}$ — derived slope, enters DC weights.

Model: Open Economy

$$\underbrace{P_t^{FH} = S_t P_t^{F*}}_{\text{LOP}}, \qquad \underbrace{(1+i_t) = (1+i^*)\Big[1-\psi(\hat{b}_t^* - \bar{b}^*)\Big]\,\mathbb{E}_t[S_{t+1}/S_t]}_{\text{UIP}}$$

$$\hat{b}_t^* = \frac{1+i_{t-1}}{\Pi_t^C}\,\hat{b}_{t-1}^* + \frac{P_t^H EX_t - P_t^{FH} IM_t}{P_t^C}, \qquad EX_t = \kappa_{EX} D_t^* \left(\frac{P_t^H}{P_t^{FH}}\right)^{-\theta^*}$$

$$Y_{it} = \beta_i^H \frac{P_t^H}{P_{it}}\left(C_t^H + EX_t\right) + \sum_j \Omega_{ji}^H \frac{MC_{jt} Y_{jt}}{P_{it}}$$

$\hat{b}^*$: NFA (scaled) EX, IM: exports, imports θ^*: export elasticity D^*: foreign demand shifter
β_i^H: cons. share, sector i Float: i_t Taylor rule, S_t free Peg: $S_t = \bar{S}$, i_t residual

Model: Policy Regimes and Shocks

Regime	Rule	Exchange rate
Float	$i_t = \bar{\imath} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$	fully endogenous (UIP)
Peg	$S_t = \bar{S}$	fixed; i_t residual
Managed	$i_t = \bar{\imath} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \log(S_t/\bar{S})$	partially stabilized

Shocks (AR(1)): a_{it} (TFP, ρ=0.90) p_t^{F*} (import price, ρ=0.85) D_t^* (foreign demand, ρ=0.80)

Theory: Regime Comparison — Predictions

	Float	**Peg**	**Managed**
Exchange rate	endogenous	fixed	partial
FX cost-push absorbed by	S_t	π_i, y	split
Monetary independence	full	none	full

Naive: float best. **Overturned** — see Welfare section.

Does the DC Index Survive a Second Cost-Push Channel?

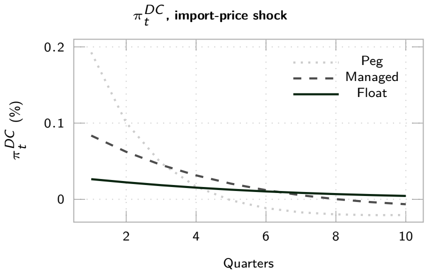
Closed economy (Rubbo): 1 channel (TFP) $\Rightarrow$ unique DC index

$$\phi^\top \mathcal{V} = \mathbf{0}^\top \Rightarrow \pi_t^{DC} = \sum_i w_i^{DC} \pi_{it}$$

Open economy: 2 channels (TFP + FX) $\Rightarrow$ need *both*, same weights ϕ

$$\underbrace{\phi^\top \mathcal{V} = \mathbf{0}^\top}_{\text{TFP}} \text{ and } \underbrace{\phi^\top \Gamma = 0}_{\text{FX}} \Rightarrow \textbf{generically infeasible}$$

- **Answer: no**, generically
- This calibration: $\cos(\Gamma, \mathcal{B}) \approx 0.96$ — close, but not constant
- **Fix:** flexible import pricing ($\hat{\delta}_M$=1) restores π_t^{DC}



Full DC-index algebra, proof, and knife-edge test: Appendix.

Calibration: 3-Sector Oil-Exporting Economy

Parameter	**Value**	**Parameter**	**Value**
β	0.99	$\Omega^H_{21}, \Omega^H_{32}$	0.20, 0.25
γ, φ	1.00, 2.00	$\Omega^F_1, \Omega^F_2, \Omega^F_3$	0.30, 0.10, 0.05
η	1.50	$\beta^H_1, \beta^H_2, \beta^H_3$	0.05, 0.15, 0.80
ε	8.00	$\delta_1, \delta_2, \delta_3$	0.75, 0.50, 0.25
θ^*	2.00	ψ	0.020
$\beta^{F,tot}$	0.10	ϕ_π, ϕ_y, ϕ_s	1.50, 0.50, 0.30

- Resource (upstream, flexible) $\rightarrow$ Manuf. (mid-chain) $\rightarrow$ Services (sticky, consumption)
- Solved nonlinear, order 1; $W^* = 1.3189$

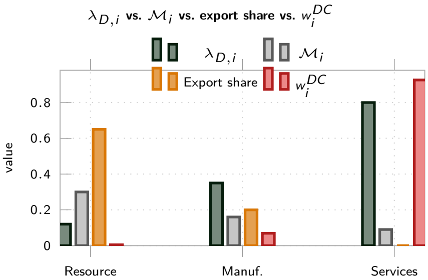
Calibration: Network Properties

$$\lambda_{D,i} = \left[(\beta^H)^\top (I - \Omega^H)^{-1}\right]_i$$

$$\mathcal{M}_i = \left[(I - \Omega^H)^{-1} \Omega^F \mathbf{1}\right]_i$$

$$w_i^{DC} \propto \lambda_{D,i} \frac{1-\hat{\delta}_i}{\hat{\delta}_i}$$

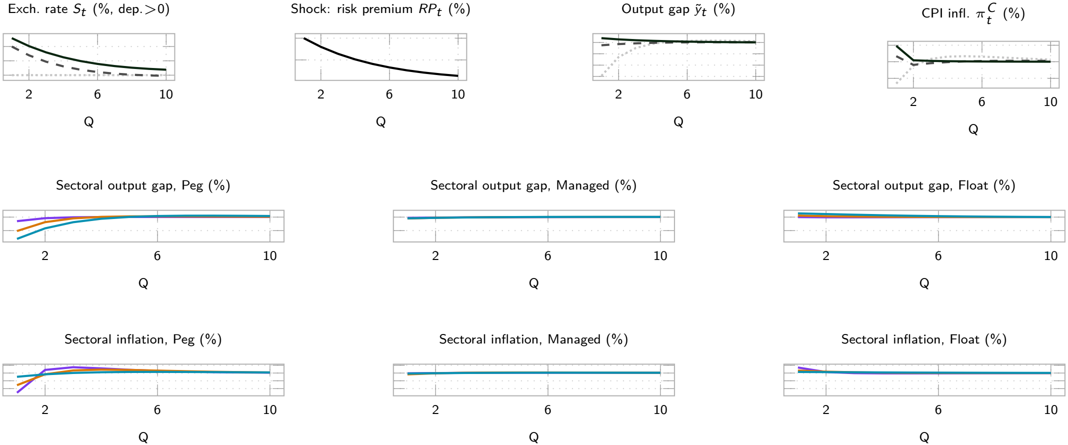
$\lambda_{D,i}$: Domar weight $\mathcal{M}_i$: import centrality w_i^{DC}: DC weight



- FX exposure runs **opposite** to Domar weight
- Resource: small weight (0.12), most FX-exposed (0.30, 0.65)
- Services: dominates output (0.80), FX-exposed only **indirectly**
- w_i^{DC} is essentially a Services index (0.93)

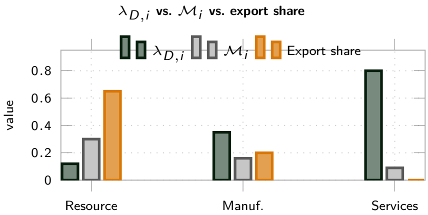
IRFs: Foreign Risk-Premium (UIP) Shock — the Key Mechanism

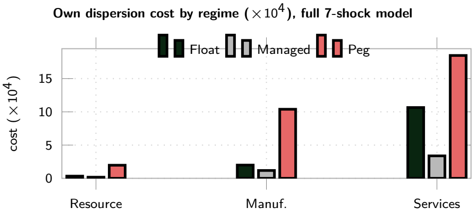
The shock that decides the ranking: 81% of Peg's welfare loss.



Peg ▪ ▪ ▪ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.

Network Exposure & Sector Welfare Preferences



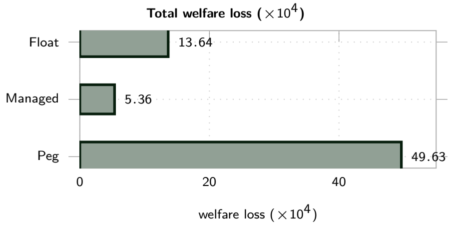


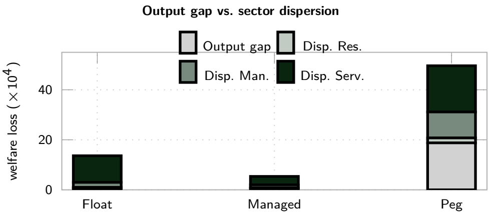
Exposure → pass-through $\Gamma_i = \frac{[\Delta(I-\Omega\Delta)^{-1}\boldsymbol{\omega}_F]_i}{\kappa_i}$ and **welfare weight** $\kappa_i = \lambda_{D,i}\varepsilon\frac{1-\hat{\delta}_i}{\hat{\delta}_i}$:

	$\mathcal{M}_i$	Export share	Γ_i	κ_i
Resource	**0.30**	**0.65**	**0.234**	0.43
Manuf.	0.16	0.20	0.054	5.54
Services	0.09	0.00	0.006	**74.6**

- With risk-premium/UIP included, **Peg no longer wins for any sector** — costliest for Resource, Manuf. *and* Services alike
- **Managed dominates every sector**; Float > Peg everywhere, most narrowly for Services

Welfare: Headline & Persistence



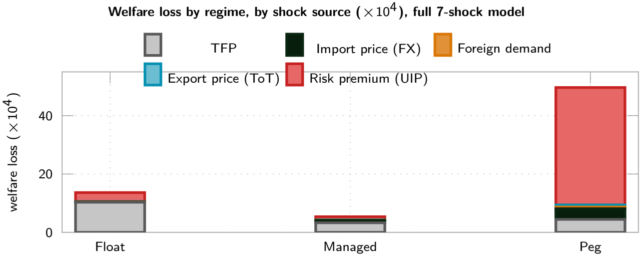


$\mathcal{W} = \frac{1}{2}\left[(\gamma+\varphi)\mathrm{Var}(\tilde{y}_t) + \sum_i \lambda_{D,i}\varepsilon\frac{1-\hat{\delta}_i}{\hat{\delta}_i}\mathrm{Var}(\pi_{it})\right]$ — **Peg dominated** on output gap & dispersion, every sector.

Why: persistence, not volatility.

- Float's S_t (near-unit-root NFA) feeds *every* sector's MC, even for domestic shocks
- Services TFP shock: Peg overshoots and reverts fast; Float barely decays
- Slow decay, not larger impact, drives Float's dispersion above Managed's

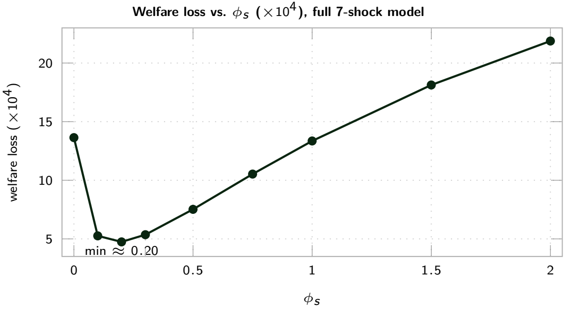
Shock Decomposition of Welfare Cost



- **Risk-premium/UIP dominates Peg** (81% of its total)
- **Ranking:** Managed $<$ Float $\ll$ Peg — Peg no longer competitive

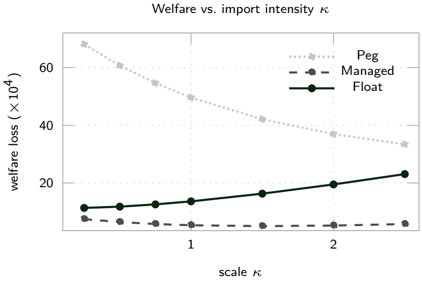
Caveat: 1st-order welfare — this channel is where 2nd-order could move most (see Conclusion).

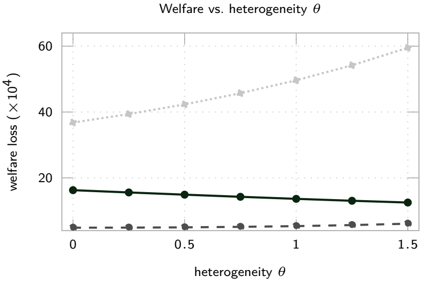
Generalization: Optimal FX Stabilization ϕ_s



- **Why interior:** too low $\rightarrow$ UIP shock leaks in; too high $\rightarrow$ loses autonomy
- ϕ_s=0 $\rightarrow$ 0.20: loss 13.64 $\rightarrow$ 4.74
- **Peaked, not flat:** 0.20 beats calibrated 0.30 by $\approx$13%
- Beyond $\approx$ 0.2: rises fast, back above Float by ϕ_s=1

Generalization: Robustness





Managed $<$ **Float** $<$ **Peg robust**, but the gap moves in **opposite directions**:

- κ (import scale): Peg falls, Float rises — **converge**
- θ (concentration): Peg rises, Float falls — **diverge**

Conclusion

- **Managed float optimal**: 5.36 vs. Float 13.64, Peg 49.63 ($\times 10^4$)
- **Driver:** risk-premium/UIP dominates Peg (81%); TFP persistence dominates Float & Managed
- **Optimal** $\phi_s \approx 0.20$: sharply peaked — over-stabilizing S_t is costly
- **Ranking robust** to κ, θ; the *gap* is not
- **Vs. literature:** same ranking as Galí–Monacelli, different channel (risk-premium, not ToT)
- **Caveat:** 1st-order welfare; UIP wedge & near-unit-root NFA are new distortions
- **Next:** order-2 solution; then δ_i, Ω^H sensitivity

Appendix: DC Index — Definition and Insulation

Closed economy (Rubbo): 1 channel (TFP), $\mathcal{V}$ rank-deficient $\Rightarrow$ unique DC index:

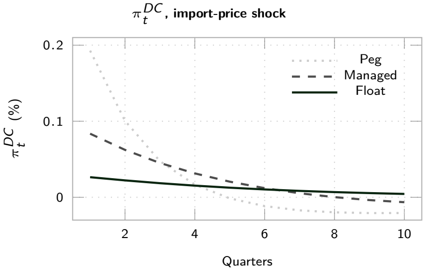
$$\phi^\top \mathcal{V} = \mathbf{0}^\top \Rightarrow \pi_t^{DC} = \sum_i w_i^{DC} \pi_{it}, \;\; w_i^{DC} \propto \lambda_{D,i} \frac{1-\hat{\delta}_i}{\hat{\delta}_i}$$

Open economy: 2 channels (TFP + FX):

$$\underbrace{\phi^\top \mathcal{V} = \mathbf{0}^\top}_{\text{TFP}} \text{ and } \underbrace{\phi^\top \Gamma = 0}_{\text{FX}} \Rightarrow \textbf{generically infeasible}$$

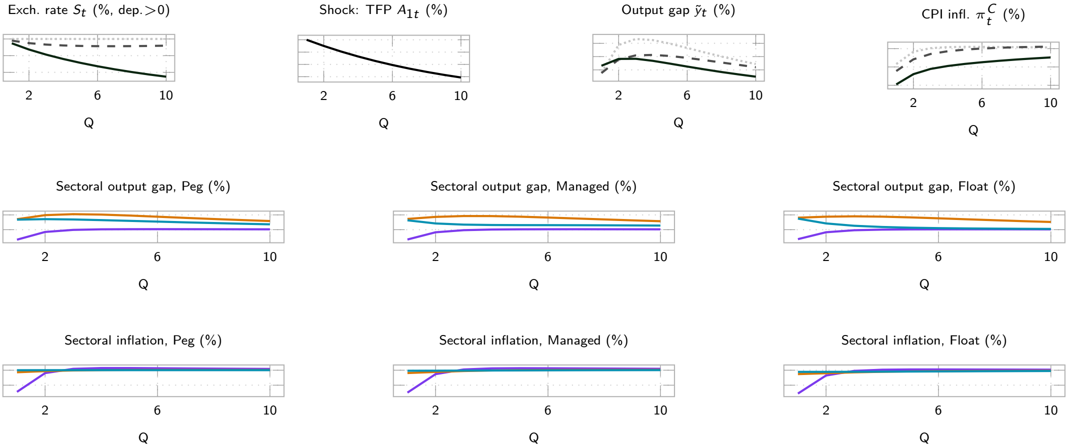
Knife-edge test, this calibration: $\Gamma \propto \mathcal{B}$ would save the DC index (Section 5.1). Instead: $\cos(\Gamma, \mathcal{B}) \approx 0.96$ ($\approx 17^\circ$ apart) — close, but $\Gamma_i / \mathcal{B}_i$ shrinks $\approx 2\times$ per sector (0.43, 0.18, 0.08), not constant. Breakdown is real, not a numerical artifact of an near-knife-edge calibration.

Fix: import nodes flexible ($\hat{\delta}_M{=}1 \Rightarrow w_M{=}0$) $\Rightarrow$ DC well-defined again by construction. This is what the Taylor rules target throughout (π_t^{DC}); FX insulation itself is a quantitative question, shown at right.



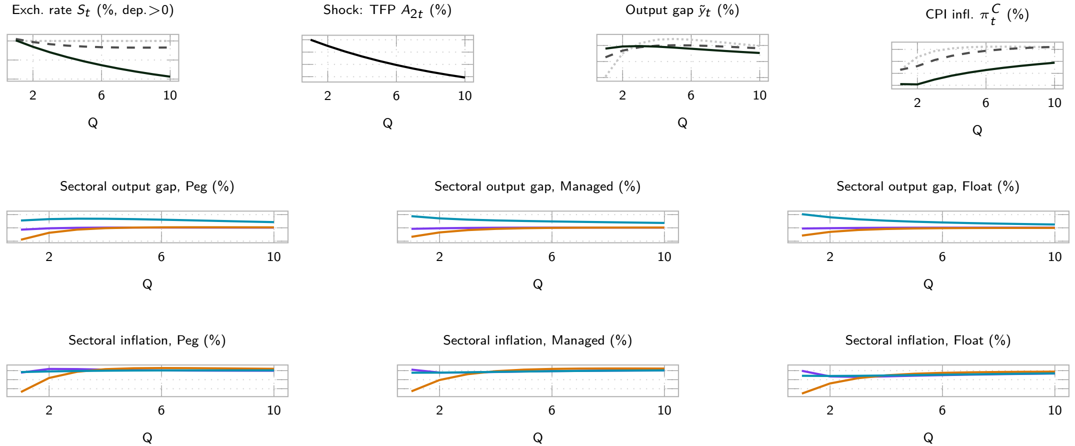
Float: smallest on impact (0.026%) & most persistent; Peg/Managed overshoot and reverse sign — on-impact insulation $\neq$ full story.

IRFs: Resource TFP Shock



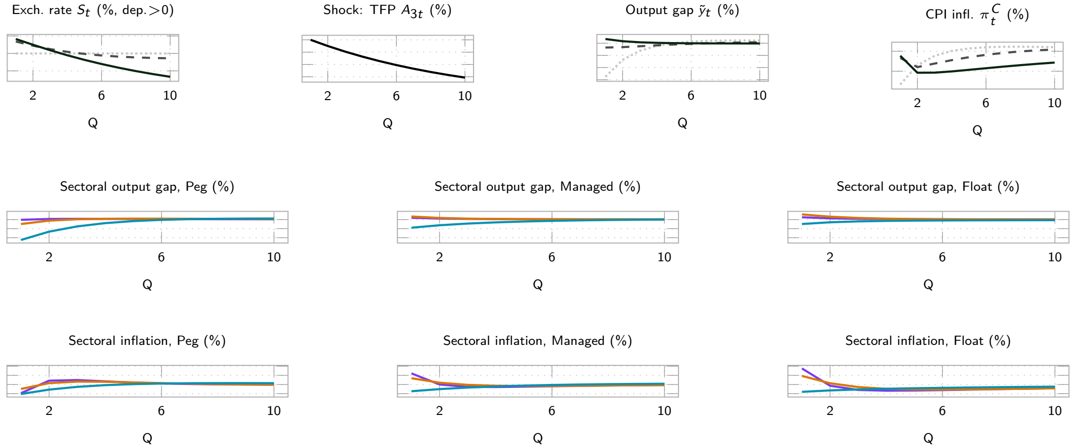
Peg ▪ ▪ ▪ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.

IRFs: Manufacturing TFP Shock



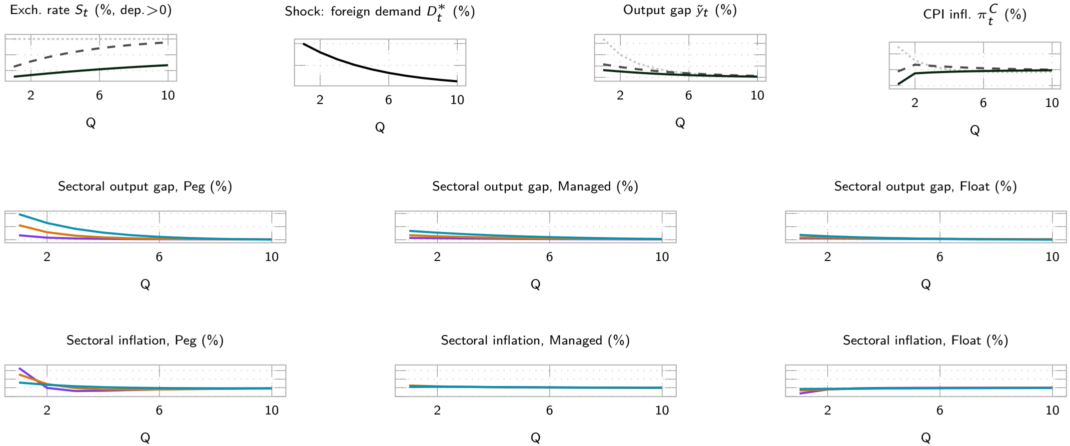
Peg ▪ ▪ ▪ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.

IRFs: Services TFP Shock



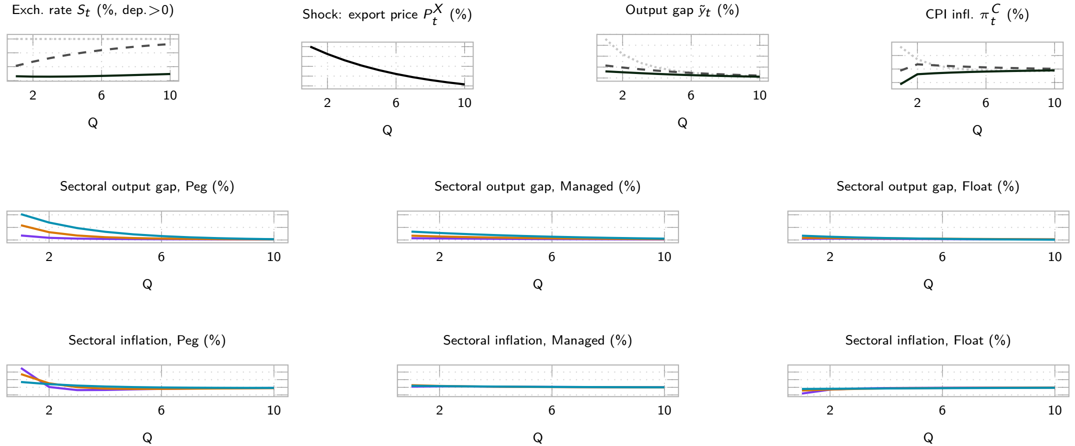
Peg ▪ ▪ ▪ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.

IRFs: Foreign-Demand Shock



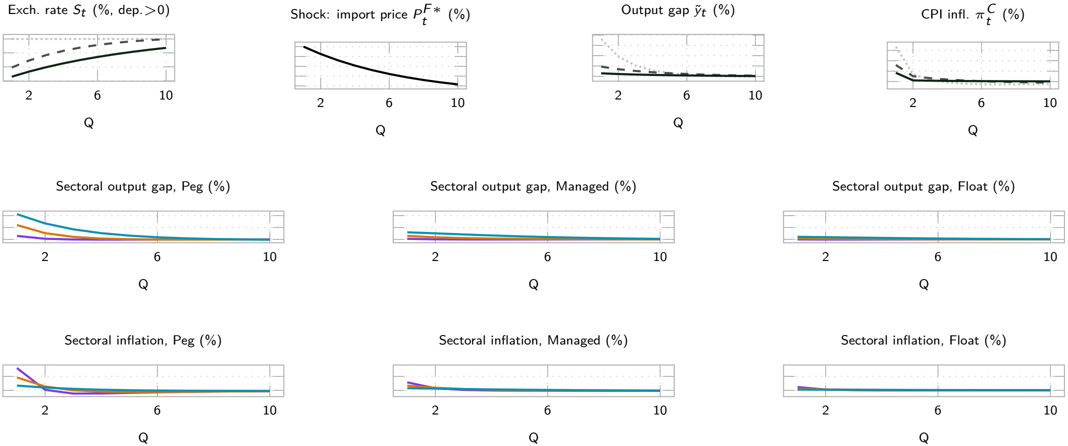
Peg ▪ ▪ ▪ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.

IRFs: Export-Price / ToT Shock



Peg ▪ ▪ ▪ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.

IRFs: Import-Price Shock (Backup)



Peg ▪ ▪ ▪ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.